

Coverage, connectivity, and topology control in sensor networks

- Efficient node deployment strategies that maximize sensing coverage and maintain connectivity under resource constraints
- Spherical Delaunay triangulation based topology control to minimize transmission power while optimizing throughput and prolonging network lifetime

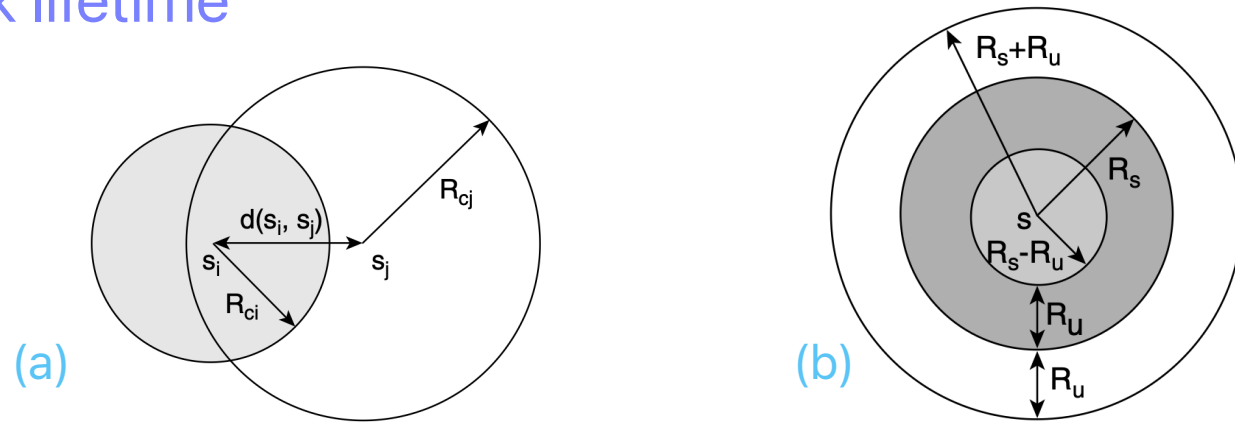


Fig 1. (a) **Communication model**: Two sensors can communicate with each other if the distance between them is $\leq$ the min of their communication radii. (b) **Probabilistic sensing/coverage model**: At distance $\leq (R_s - R_u)$, coverage is 1; at distance $\geq (R_s + R_u)$, coverage is 0; within $(R_s - R_u)$ and $(R_s + R_u)$, coverage decays exponentially.

Fig 2. **Minimum connected-sensor cover problem**: To find a minimum set of nodes that can provide complete coverage of the region while also guaranteeing a connected network. Nodes denoted by black dots comprise such a set.

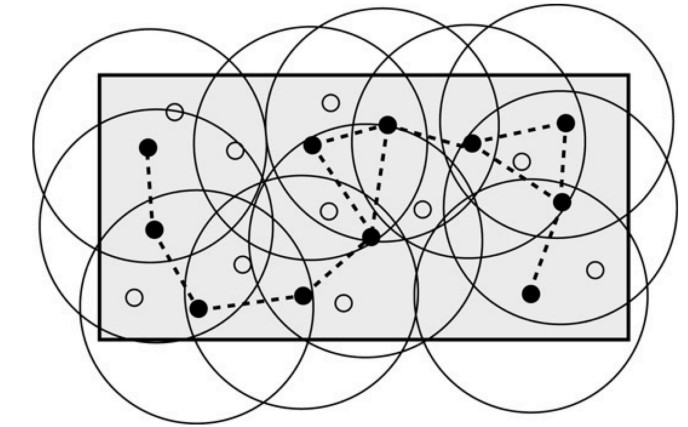


Fig 3. **Spherical Delaunay triangulation** - a dual of Voronoi diagrams - of 100 points randomly distributed on the surface of a sphere.

A Voronoi diagram for a set of N points in 2D or 3D tessellates the 2D or 3D region into a set of N convex polygons or polyhedra such that any point lying within a polygon or polyhedron is closest to the point that is inside the polygon or polyhedron.

